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EXPT NO. - 12

TITLE - STACK USING ARRAYS

```
In [10]: def push(stack, item, max_size):
    if len(stack) >= max_size:
        print("Stack Overflow")
    else:
        stack.append(item)
        print("Pushed", item, "to stack")

def pop(stack):
    if not stack:
        print("Stack Underflow")
    else:
        item = stack.pop()
        print("Popped", item, "from stack")
        return item

def display(stack):
    if not stack:
        print("Stack is empty")
    else:
        print("Stack elements are:")
        for item in reversed(stack):
            print(item)

# Input
max_size = int(input("Enter the maximum size of the stack: "))
stack = []

while True:
    print("=====")
    print("\n1. Push\n2. Pop\n3. Display\n4. Exit")
    choice = int(input("\nEnter your choice: "))

    if choice == 1:
        item = int(input("Enter the item to push: "))
        push(stack, item, max_size)
    elif choice == 2:
        pop(stack)
    elif choice == 3:
        display(stack)
    elif choice == 4:
        break
    else:
        print("Invalid choice, please try again...")
```

=====

1. Push
2. Pop
3. Display
4. Exit

```
Pushed 10 to stack
=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 20 to stack
=====

1. Push
2. Pop
3. Display
4. Exit
Stack elements are:
20
10
=====

1. Push
2. Pop
3. Display
4. Exit
Popped 20 from stack
=====

1. Push
2. Pop
3. Display
4. Exit
Stack elements are:
10
=====

1. Push
2. Pop
3. Display
4. Exit
Popped 10 from stack
=====

1. Push
2. Pop
3. Display
4. Exit
Stack is empty
=====

1. Push
2. Pop
3. Display
4. Exit
Stack Underflow
=====

1. Push
2. Pop
3. Display
4. Exit
Invalid choice, please try again...
=====

1. Push
2. Pop
3. Display
4. Exit
```